

LECTURE L1-2 • MODULE 1 • CO1

The role of EDA, and robust statistics

Week 1 | Slides: DATA209_Module_1_Statistical_and_Data_Foundations.pptx, pages 4-19 | Laboratory: P1-2

TOPICS COVERED

- Role of EDA in the data science lifecycle
- Exploratory vs. confirmatory analysis
- Sampling bias and inference intuition
- Distribution behaviour: skewness, kurtosis, outliers
- Summary and robust statistics — median, IQR

What EDA is for

Exploratory data analysis is the practice of looking at what the data *seems* to say before any hypothesis is imposed on it. Tukey named the field in 1977 as a deliberate counterweight to a statistical culture that had become almost entirely confirmatory. His argument was that the analyst's first duty is detective work — finding what is there — and that this work has its own methods, standards and failure modes.

The practical case is simpler. You cannot draw reliable conclusions from a large quantity of data by gleaning over it. Raw data is skewed, contains impossible values, and hides its own collection history. A model fitted to unexamined data inherits every one of those defects and reports them with false confidence.

Exploratory versus confirmatory

	EXPLORATORY	CONFIRMATORY
Question	open — what is going on here?	fixed before the data is seen
Hypothesis	an output	an input
Method	graphical, iterative, sceptical	formal tests, error rates
Success	a surprising, checkable claim	a defensible accept/reject decision

The cardinal error is switching modes silently: exploring until something looks significant, then reporting it as though the hypothesis came first. That is p-hacking, and it is the reason EDA findings are stated as *hypotheses for future testing* rather than as results.

Sampling and what you may claim

Every statistic is an estimate from a window onto a population. Four ways that window distorts:

- **Selection bias** — the sampling mechanism favours certain units. Website analytics only observe people who reached the website.
- **Survivorship bias** — failures are absent. Loan data holds approved applicants; rejected ones

are invisible.

- **Non-response bias** — those who did not answer differ systematically from those who did.
- **Measurement bias** — the instrument distorts. A sensor drifts; a zero means "not recorded".

Larger samples reduce variance, **not bias**. A biased frame stays biased at any size. Before generalising, ask the question that settles most disputes: *who could have been in this dataset but is not?*

Summary statistics: position, dispersion, shape

Three families of summary, and the discipline of pairing them correctly.

- **Position** — mean (balance point), median (splits the data in half), mode (most frequent).

The mean is not wrong on skewed data; it answers a different question. The size of the mean–median gap is the diagnostic.

- **Dispersion** — range, variance, standard deviation, IQR. The first three are computed around

the mean and inherit its sensitivity to extremes. The IQR is the width of the middle 50% and ignores the tails entirely.

- **Shape** — skewness measures horizontal asymmetry; kurtosis measures tail weight.

Report **mean with standard deviation**, or **median with IQR**. Never mix a mean with an IQR.

On `pima.csv`, Insulin has skew +2.27; its mean is 155.5 against a median of 125.0. Both numbers are correct. Only one describes a typical patient.

Reading a boxplot

The box spans Q1 to Q3 with the **median** — not the mean — inside it. Whiskers extend to the furthest point within $1.5 \times \text{IQR}$ of the box edges; points beyond are *flagged*, not condemned. Box position within the whiskers reveals skew: if $Q3 - Q2$ exceeds $Q2 - Q1$, the distribution leans right. A boxplot hides multimodality completely, so always pair it with a histogram or density plot.

KEY TERMS

Exploratory data analysis Confirmatory analysis Sampling frame Selection bias
Survivorship bias Robust statistic Skewness Kurtosis Interquartile range
Five-number summary

COMMON ERRORS TO AVOID

- Reporting a mean for a strongly skewed variable without also giving the median
- Pairing a mean with an IQR, or a median with a standard deviation
- Generalising beyond the sampling frame without saying so
- Treating flagged boxplot points as errors before investigating them

1. NIST/SEMATECH e-Handbook — Ch.1 Exploratory Data Analysis
<https://www.itl.nist.gov/div898/handbook/eda/eda.htm>
2. Downey — Think Stats, 2nd ed. (Ch. 1–4)
<https://greenteapress.com/wp/think-stats-2e/>
3. Adhikari & DeNero — Computational and Inferential Thinking
<https://inferentialthinking.com/>
4. MIT OCW 6.0002 — sampling and statistical fallacies
<https://ocw.mit.edu/courses/6-0002-introduction-to-computational-thinking-and-data-science-fall-2016/>

TEST YOURSELF

1. Give a variable for which the mode is the only meaningful measure of position.
2. A dataset has mean 40 and median 12. Sketch the distribution and name the summary you would report.
3. Why does increasing the sample size not fix selection bias?
4. $Q_3 - Q_2 = 5$ and $Q_2 - Q_1 = 18$. Which way does the distribution lean?